FLEET TEAM DIGEST

737MAX-FTD-34-18001

Issue Title : Performance Based Communication and Surveillance (PBCS)

Airplane Model	ATA	Minor Model(s)
737MAX	3400-00	-7, -8, -8200, -9, -BBJ

Other Model(s)	ECCN
757, MD-11, 767, 777, 717, 747, 737NG, 787	9E991

Originated Date	Last Revised Date	Created On
01/09/2018	06/22/2018	06/22/2018

Estimated Completion Date	Next Update Date

Revision Description

Status, Final Action and Operator Action updated.

Description

Operators desiring to use Performance Based separation standards beginning on 29 March 2018 and beyond will need to meet the requirements for RCP240 and RSP180. North Atlantic System Planning Group (NAT SPG) and Asia Pacific Air Navigation Planning and Implementation Regional Group (APANPIRG) regions will begin implementation of Performance Based Communication and Surveillance global requirements in starting 29 March 2018.

Compliance has been demonstrated for:

-787

Many airlines in the IATA and NAT region have asked Boeing to address the compliance issue of Boeing models which at this point have NOT demonstrated PBCS RCP/RSP compliance.

-737 with FMC U13 or earlier software versions

-737 with FMC U14 software - certification planned for 1Q2019.

-747-400 with legacy FMC hardware does NOT have the latency timer feature therefore will NOT meet the PBCS operational requirements.

-747-400 with BP4.0 NG FMC - Certification planned for 1Q2019.

-747-8 with BP4.0 - Certification planned for 1Q2019.

-757/767 with Peg II - Certification planned for 2Q2018.

-777 does not pass Satcom latency timer requirements.

-MD11

-717

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Background

PBCS Provides a framework for managing communication and surveillance performance in accordance with globally accepted Required Communication Performance (RCP) and Required Surveillance Performance (RSP) specifications.

The RCP/RSP specifications provides the means to:

-Initially qualify different sub-systems , i.e. Aircraft, Network and ATS Unit

-Manage operational system performance (end-to-end) in continued operations.

ICAO developed Annexes 6 and 11 and the Procedures for Air Navigation Services (PANS-ATM) Doc 4444 adopted provisions for PBCS that became applicable in November 2016.

-Doc 9869, Performance-based Communication and Surveillance (PBCS) Manual, Edition 2

-Doc 10037, Global Operational Data Link (GOLD) Manual, Edition 1

As of September 2016, airspace using or planning to use PBCS RSP/RSP approvals are:

-NAT FIRs: Gander, Shanwick, Reykjavik, New York, Santa Maria

-PAC FIRs: Anchorage, Auckland, Brisbane, Fukuoka, Nadi, Oakland, Port Moresby, Tahiti

-Asia: Some routes over Bay of Bengal and South China Sea.

To support airline operational approval, Boeing is planning to demonstrate RCP240/RSP180 compliance for those models/ systems that have not been demonstrated

-Lab testing will have to be done to collect the Transaction Time and Continuity

-ATS System Requirements & Objectives (SR&O) will have to be revised to address the Availability, Integrity and Safety Requirements (including the collected Transaction Time and Continuity)

-The AFM will not be revised since it already referred to ATS SR&O

PBCS is only required for oceanic environments at this time. The North Atlantic and the South East Asia regions are the first two with the mandate starting March of 2018.

Status

Boeing has approved an internal project to begin testing and update ATS SR&O.

There are many different datalink configurations in use by the Operators. Boeing is trying to determine the specific datalink configurations. The plan is to test in two phases. Phase 1 Configurations, Phase 2 Configurations.

Phase 1 Configurations evaluation Plan is complete.

See PHASE 1 Configurations attachment.

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Phase 2 Configurations evaluation Plan will be based on Operator feedback from MOM-MOM-17-0716 responses.

MOM was released.

BPCS Phase 1 Test Results:

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-737 NG/MAX with A4 HW with U12 with RC (822-1239-151) PASSED PBCS requirements.

-737 NG/MAX with A4 HW with U12 with HI (965-0758-006) FAILED PBCS requirements.

-737 NG/MAX with C1 HW with U13 with RC (822-1239-151) PASSED PBCS requirements.

-737 NG/MAX with C1 HW with U13 with HI (965-0758-006) FAILED PBCS requirements.

Boeing has coordinated with the supplier, Honeywell, to address the configurations that failed.

This effort will be managed separately and scheduled is TBD.

-Phase 1 Configuration ATS SR&0 was updated/published by on 28 Feb 2018 for those configurations that passed testing.

-Phase 2 has complete as of JUNE2018.

-The ATS SR&O has been updated to include the successful configurations using VDL and SATCOM (INMARSAT and Iridium) subnetworks.

-The testing was accomplished using the Rockwell Collins CMU with Core SW P/N 832-9548-012 with following FMC SW P/N:

556909-001 (FMC U11.0)

559775-001 (FMC U12.0)

559777-001 (FMC U13.0)

Interim Action

Operators affected by the PBCS mandate should initiate dialogue with their authority counterparts to assure there is a mutual understanding of PBCS approvals supporting documents.

Final Action

The 737 PBCS Phase 1 (MARCH2018) and Phase 2 (JUNE2018) has completed.

Boeing has updated with the ATS SR&O document and can provide to Operators to use upon request.

Operator Action

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Operators can make a request for the updated ATS SR&O document available now via BCS Service Request.

Reference Categories

Reference Type	Reference Number	Revision Number	Reference Date
Multi-Operator Message (MOM)	MOM-MOM-18-0129-01B		03/14/2018
Multi-Operator Message (MOM)	MOM-MOM-17-0716-01B		12/21/2017

Related Categories

- Aging Airplane Issues : Structural issues or systems wear due to high flight cycles or high flight hours on the airplane
- Regulatory : Issues mandated by airworthiness directive or other regulatory guidance material

Attachments:

a) PHASE 1 Configurations Rev1

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